Do the Judge the position factors tell who is better?

Five board readings against Stockfish, with material level, in 10-minute games at the player's band

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Abstract

The app's Judge the position drill explains who is better with factors its board code scores in centipawns. With material level, does each factor beyond material, and the worst piece, point to the side Stockfish says is better? In 21,266 positions from Lichess games at 10 minutes between players at the player's band and the band above, all five do on both months, so by the rule fixed before the run each *tells who is better*. Ranked by held-out agreement: something loose (92.9%), activity and space (82.3%), king safety (81.2%), the worst piece (74.2%), then pawn structure (68.9%). All but pawn structure are right at least 67% of the time on their own. Weighed together by a logistic model, the readings name the better side in 85.6% of the held-out positions with a clear verdict. Added after the run: whose move it is matters too. Where nothing is loose, the side to move is the better side in 66% of clear positions. A reading that favours the side to move is right 90% to 98% of the time, while pawn structure and the worst piece, when they favour the side that just moved, are no better than a coin. When two readings disagree, a loose pawn usually wins and pawn structure usually loses.

1 The question

A coach judges a position before choosing a move: count the material, then look at the kings, the pieces and the pawns. The app's Judge the position drill teaches that habit. It shows a position from the player's game, asks who is better, and explains the engine's answer with five factors the board code scores in centipawns: material, king safety, activity and space, pawn structure, and something loose. The drill keeps a card only when the largest factor agrees with the engine, so on its own cards a factor is right by construction. What its cards cannot show is whether a factor points to the better side in the positions he will meet, which is what the habit needs.

This study asks it of each factor beyond material, and of a sixth reading the app teaches elsewhere, the worst piece, in positions where material is level. A reading that points to the better side is worth teaching as a reason. One that does not would teach him to trust a sign that tells him nothing, and one that points the wrong way is a bug in the board code.

2 How it was measured

The readings. All are the app's, at commit `f4bd63f19b`, each from White's side (positive when it favours White):

- **King safety**: each king's danger is 25 for each pawn missing from its shelter, 12 for each point of weight of the enemy pieces attacking the squares around it, and 30 for each open file beside it (20 if half open); Black's danger minus White's, scaled down as the pieces come off.

- **Activity and space**: 4 for each legal move more than the other side (counting each side's moves as if it were to move), and 4 for each square more it attacks in the other side's half.
- **Pawn structure**: each side loses 15 for a doubled pawn and 20 for an isolated one, and gains 20 for a passed pawn and 12 for each rank it has advanced past its third, the passers counting up to twice as much as the pieces come off; White's sum minus Black's.
- **Something loose**: the best capture balance of the side to move, less part of the other side's, each counted only when it wins material. In these positions, where neither side can win more than one pawn by a capture, it leans only when the side to move can win a pawn and the other side has nothing to take, and it then favours the side to move.
- **The worst piece**: the legal moves of White's least mobile piece (not a pawn or the king) minus those of Black's. "Feed your laziest piece" is one of the app's lessons; the drill folds it into activity.

A reading *leans* when it is at least the drill's 80 centipawns, or three moves for the worst piece, and it favours the side its sign names.

The games. Lichess's rated games (the database's August 2026 file for training and September 2026 held out) at 10 minutes with no increment, the player's time control, that ended normally or on time after at least 30 plies, with both players rated 1250 to 1649: his band on Lichess (1250–1449) and the band above (1450–1649). The first 8,000 such games of each month enter.

The positions. The drill's own population: after at least 12 plies, not over and not in check, with at least four legal moves, material exactly level, and no tactic (no mate in one for either side, and no capture that wins more than one pawn for either). Either side may be to move. Of the 470,133 positions after the first 12 plies in the train games, 103,668 (22%) qualify, and 62% fail because material is not level. The held-out games give 105,341 of 474,663 (22%). For each reading, a game gives at most one position: one drawn at random, seeded by the game and the reading, among its positions where the reading leans. Each reading keeps up to 3,000 positions a month, in file order: 23,253 rows over the five readings, 21,266 distinct positions.

The verdict. Stockfish through the app's engine service at the drill's grading budget (250,000 nodes), with its hash cleared before each position and no tablebase. On the drill's own scale, a position is *level* within 50 centipawns and *clear* at 150 or more; a mate is clear. A reading's *agreement* is the share of its positions with a clear verdict where the side it favours is the side the engine says is better. Each position comes from a different game, so its interval is the 95% Wilson interval.

The rule, fixed before any position was scored. For each reading, on each month separately: it *tells who is better* when its agreement's interval lies above 50% on both; it is *reliable on its own* when the interval lies above 67% on both; it *points the wrong way* when the interval lies below 50% on both; and there is *no signal* otherwise. Under 200 held-out positions with a clear verdict, the verdict is *not enough data*. Material is level in every position, so a reading that tells who is better adds to material by construction.

		Train		Held out			
Reading	Leans at	Clear	Agreement, %	Clear	Agreement, %	Verdict	Reliable
King safety	80 cp	747	77.2 (74.1–80.1)	732	81.2 (78.1–83.8)	tells who is better	yes
Activity and space	80 cp	776	81.7 (78.8–84.3)	826	82.3 (79.6–84.8)	tells who is better	yes
Pawn structure	80 cp	384	63.0 (58.1–67.7)	405	68.9 (64.2–73.2)	tells who is better	no
Something loose	80 cp	1,409	92.6 (91.1–93.9)	1,388	92.9 (91.5–94.2)	tells who is better	yes
The worst piece	3 moves	1,055	71.5 (68.7–74.1)	1,023	74.2 (71.4–76.8)	tells who is better	yes

Table 1: Each reading's agreement with the engine: of its positions with a clear verdict (*Clear*), the share where the side the reading favours is the side Stockfish says is better, with the 95% Wilson interval. The verdict, and whether the reading is reliable on its own, are the rule's, fixed before any position was scored.

3 The verdict

All five readings tell who is better (Table 1). On both months each agreement's interval lies above 50%, and the readings rank in the same order on both: something loose (92.9%), activity and space (82.3%), king safety (81.2%), the worst piece (74.2%), then pawn structure (68.9%) held out. All but pawn structure are also reliable on their own, with intervals above 67%; pawn structure falls short of that bar on both months. No reading points the wrong way, so the board code has no factor whose sign is backwards.

4 What the positions describe

	Engine's verdict		Band		Phase			
Reading	Level	Clear	1250–1449	1450–1649	Opening	Middle	End	Twice the lean
King safety	30	38	81 (340)	82 (392)	78 (136)	82 (541)	82 (55)	97 (32)
Activity and space	37	28	80 (389)	84 (437)	79 (248)	84 (473)	84 (105)	– (18)
Pawn structure	25	53	65 (182)	72 (223)	– (6)	65 (122)	71 (277)	69 (49)
Something loose	20	46	94 (621)	92 (767)	92 (338)	93 (737)	94 (313)	– (0)
The worst piece	35	34	75 (449)	74 (574)	– (11)	76 (630)	71 (382)	74 (148)

Table 2: Held-out positions only. *Level* and *Clear*: the share of a reading's positions the engine calls level or clear. The other columns give the agreement, with the number of clear positions in brackets, by the band of the two players' mean rating, by the divider's phase, and where the reading is at least twice its lean; a cell with fewer than 30 clear positions shows only its count. These were named before the run and carry no rule.

A lean is a sign, not a verdict. The engine calls between 20% and 37% of a reading's held-out positions level (Table 2). The size of a lean matters for the kings and the pieces: at twice its lean, king safety agrees in 97% of 32 held-out positions and activity in 94% of 18, though these are few. Pawn structure and the worst piece gain nothing at twice their lean. Something loose never reaches it here, since no capture wins more than a pawn.

Reading	Decisive games	Reading's side won	Engine's side won
King safety	700	64 (60–67)	69 (65–72)
Activity and space	790	64 (61–68)	70 (67–73)
Pawn structure	375	65 (60–69)	78 (74–82)
Something loose	1,318	67 (64–69)	70 (67–72)
The worst piece	975	63 (60–66)	72 (69–75)

Table 3: Held-out positions with a clear verdict from games that ended in a win: how often the side the reading favours won the game, beside how often the side the engine favours won it, with 95% Wilson intervals.

The result. In the decisive games, the side a reading favours won 63% to 67% of the time, on both months less often than the side the engine favours (Table 3). Pawn structure trails the engine most: its side won 65% of these games, the engine's 78%.

Together. The drill adds its factors as they come, in centipawns. That plain sum favours the engine's side in 83.8 (82.6–84.9)% of 3,813 held-out positions (82.5 (81.3–83.7)% on train). A logistic model fitted on the train month, which weighs each reading, does a little better: 84.5% on train and 85.6% held out. Per 100 centipawns it weighs something loose most (2.41 in log-odds), then activity (1.48), king safety (0.96) and pawn structure (0.52). The worst piece, at 0.06 a move, moves the model less at its lean than any drill factor does at its own, so beside the other four it adds little.

5 Whose move it is

		Agreement, %		
Reading	Favours the mover	For the mover	For the other side	Only sign
King safety	42	96 (93–97)	61 (55–66)	78 (74–82)
Activity and space	38	98 (96–99)	65 (60–69)	82 (79–85)
Pawn structure	47	90 (85–93)	45 (38–52)	75 (69–80)
Something loose	100	93 (91–94)	–	95 (94–97)
The worst piece	43	93 (91–95)	51 (47–56)	74 (69–78)

Table 4: Added after the run; held-out positions. *Favours the mover*: the share of a reading's positions where it favours the side to move. The agreement, with 95% Wilson intervals, when it favours the side to move, when it favours the side that just moved, and when no other drill factor leans.

These descriptions were added once the verdicts were read; none was named before the run. They change how the agreement above should be read.

The move. Where nothing is loose, the side to move is the engine's better side in 65.8 (63.7–67.8)% of the 2,074 clear held-out positions (64.1 (62.0–66.1)% on train). With material level, a clear edge at this level is usually something the side to move can act on at once. Every reading but something loose favours the side that just moved more often than the side to move (Table 4), so for them the

move works against agreement, not for it: the move alone would put each below half. Something loose always favours the side to move, and agrees far more often than the move alone would.

For the mover and against. A reading that favours the side to move agrees 90% to 98% of the time. One that favours the side that just moved still agrees more often than not for king safety (61%) and activity (65%), but pawn structure (45%) and the worst piece (51%) are then no better than a coin on either month. When it is the only drill factor that leans, every reading agrees well, pawn structure included (75%).

		Train		Held out	
First	against	n	Sides with the first	n	Sides with the first
King safety	activity and space	33	36 (22–53)	32	44 (28–61)
King safety	pawn structure	19	79 (57–91)	9	56 (27–81)
King safety	something loose	83	40 (30–51)	88	36 (27–47)
King safety	the worst piece	66	56 (44–67)	55	64 (50–75)
Activity and space	pawn structure	34	82 (66–92)	24	75 (55–88)
Activity and space	something loose	111	35 (27–44)	96	34 (26–44)
Activity and space	the worst piece	9	89 (56–98)	8	75 (41–93)
Pawn structure	something loose	114	15 (10–23)	101	19 (12–28)
Pawn structure	the worst piece	138	49 (41–58)	120	50 (41–59)
Something loose	the worst piece	163	77 (70–83)	166	81 (75–87)

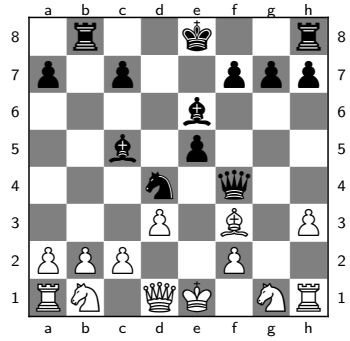
Table 5: Added after the run. The distinct positions with a clear verdict where both readings lean, in opposite directions: the share where the engine sides with the first, with 95% Wilson intervals.

When two disagree. Where two readings lean opposite ways (Table 5), the engine sides with something loose over activity (66%), pawn structure (81%) and the worst piece (81%) on both months, and over king safety (64%) on the held-out month; and with activity over pawn structure (75% of 24). Pawn structure and the worst piece split their disagreements. The other pairs are too few or too close to call.

The move as a reading. With the side to move as a sixth input, the logistic model's accuracy rises to 86.1% on train and 87.7% held out, and the move weighs more than king safety at its lean and less than activity at its lean.

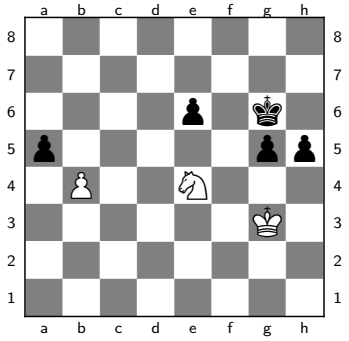
6 Two positions

Two held-out positions show a reading at work and the move outweighing one (Figure 1). They illustrate; they test nothing.



FEN `1r2k2r/p1p2ppp/4b3/2b1p3/3n1q2/3P1B1P/PPP2P2/RN1QK1NR w KQk - 0 13`

(a) White to move, in the middlegame. Only activity and space leans: 204 centipawns for Black, who has 61 legal moves to White's 25 and attacks 20 squares in White's half to 5 in Black's. Material is level and no capture wins more than a pawn, yet the engine makes Black 3.9 pawns better: 13. b3 e4 14. ♗xe4 ♘xb3.



FEN `8/8/4p1k1/p5pp/1P2N3/6K1/8/8 w - - 0 36`

(b) White to move, in the endgame. Only pawn structure leans: 211 centipawns for Black, who has three passed pawns to none. But each side has a pawn to take and White takes first, so the engine makes White 5.9 pawns better: 36. bxa5 ♔f5 37. a6 h4+.

Figure 1: Held-out positions. Left: of the middlegame positions where activity and space is the only drill factor that leans and the engine agrees with it, the one where it leans most. Right: of the positions where pawn structure is the only drill factor that leans, favouring the side that just moved, and the engine says the side to move is clearly better, the one where it leans most. Each line is the engine's, at the grading budget.

7 What the app can take

- **Keep the drill's factors.** Each points to the better side at his level, and none points the wrong way.
- **Teach pawn structure as a tie-breaker.** On its own it is usually right, but when a loose pawn or the pieces say otherwise the engine sides with them, and when it favours the side that just moved it is no better than a coin.
- **Name whose move it is.** The drill always shows a position with him to move, and the deck drops a card whose largest factor disagrees with the engine. Where the factors are quiet or favour his opponent, the engine often still says he is better because it is his move. The move as a sixth reason is a candidate for the drill, to be tested on his own cards with its rule fixed first.
- **The worst piece is mostly activity.** Beside the four factors it adds little, so the drill is right to fold it in.

8 Caveats

- **Not his games.** The positions are Lichess players' at his band and the band above; the drill's cards come from his own games.
- **The engine's verdict, not his.** Whether a reading helps him choose moves is a later study's question, on his own answers.

- **One reading at a time.** The readings move together, so a reading's agreement says it points the right way, not that it adds to the others. The logistic models describe that, and only describe.
- **Added after the run.** The side to move, the readings alone and the disagreements were looked at once the verdicts were in. They describe these positions; a rule built on them needs positions this study did not see.
- **Pawn breaks** are a plan rather than a standing edge, so they are tested as a rule of thumb, not here.